



Biomedical applications of acoustically responsive phase shift nanodroplets: Current status and future directions

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ABSTRACT

The evolution of ultrasonic contrast agents to enhance the reflectivity of structures in the human body has consolidated ultrasound's stance as a reliable diagnostic imaging modality. A significant development within this field includes the advent of liquid nanodroplets that are capable of vaporising into gaseous microbubbles upon ultrasonic irradiation. This literature review will therefore appraise and summarise the available literature on the generation of phase-shift nanodroplets, their formulations, applications, safety issues, future developments and any implications that may inhibit their clinical implementation. The main findings of this review affirm that phase change nanodroplets do indeed demonstrate functionality in drug delivery and targeting and characterisation of tumours. Its bioeffects however, have not yet been extensively researched, prompting further exploration into how bubble size can be controlled once it has vaporised into microbubbles and the resulting complications. As such, future research should be directed towards determining the safety, longevity and suitability of phase-shift nanodroplets over contrast agents in current clinical use.

1. Introduction

Since ultrasound was first used as an imaging tool for clinical purposes more than 60 years ago, its diagnostic and therapeutic applications have rapidly evolved to deliver more ways of acquiring improved images [1]. One such example is the conception of Ultrasound Contrast Agents (UCAs) in 1968 to enhance the ultrasound signal of low contrast regions [2]. Despite each type of agent having individual characteristics, uses, advantages and limitations, they all share a common purpose. This is to use their highly echogenic properties to safely and non-invasively create a strong echo for even anatomical regions with naturally low ultrasound contrast, such as blood [3]. This aids diagnosis and improves treatment through better evaluating blood flow, myocardial perfusion and provides functional and molecular tissue information [4,5].

As with progress made in improving ultrasound techniques, there has also been substantial development in the types of contrasts invented. The first sort were large suspensions of microbubbles composed of air with a shell of galactose microparticles such as Levovist® [6]. Combined with a relatively large diameter between one to ten micrometres and short half-life, these gas bubbles undergo rapid destruction following ultrasound exposure and are considered moderately unstable [2]. The culmination of these factors limits both its circulation and scan time [7].

Overcoming restrictions in these original microbubble prototypes necessitated a variation that prioritised stability with increased longevity. Thus, another form of contrast agent was manufactured. Though still termed microbubbles, they are instead composed of a more inert and slowly diffusing gas such as sulphur hexafluoride, with SonoVue® being an example [8]. Surfactants, albumin or phospholipids constitute the bubble's shell [9]. The primary aim of these microbubbles is to target vascular endothelial markers of disease by attaching biomolecules to their surface, [3] however the diameter of these microbubbles meant there was difficulty in extravasating beyond the vessels in solid tumours leading to microbubble retention [2] (Fig. 1).

Notwithstanding the advances in making contrast agents more applicable to ultrasound and the body, traditional contrast agents are still too large to pass through the vascular space [3]. Hence, the most recent group of agents to have evolved are referred to as nanodroplets which have a perfluorocarbon (PFC) liquid core stabilised by a shell of lipids, polymers or proteins [10–12]. They are in the order of 400–800 nm [2] which is a promising property for targeted molecular imaging and drug delivery to tumours as they will be able to extravasate and accumulate due to the enhanced permeability and retention effect (EPR) [13]. Other potential clinical uses include gene delivery, thermal ablation and diagnostic and therapeutic purposes. There are many forms of nanoparticles that can accomplish these applications such as polymeric micelles, liposomes, nanospheres, nanocapsules and nanoemulsion

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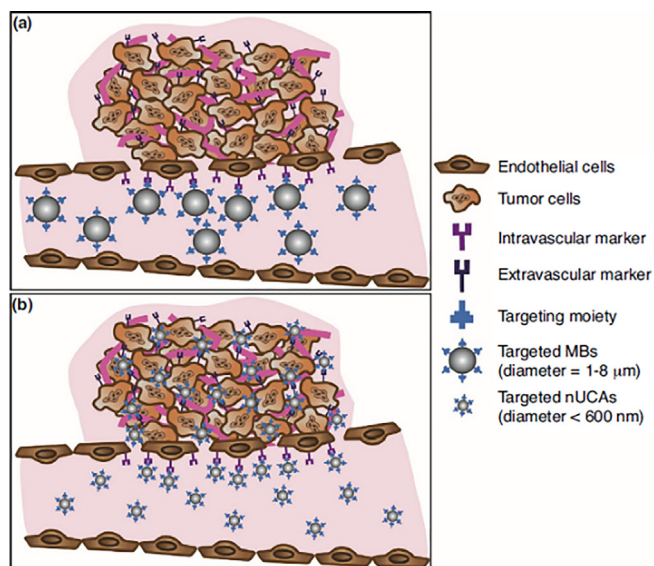


Fig. 1. Graphic illustration comparing ultrasound imaging with microbubbles and nanoparticles. (a) Imaging with targeted microbubbles that can only target intravascular markers as they are limited to the vascular space and unable to extravasate through the endothelial cell wall. (b) Imaging with targeted nanoparticles which are able to target extravascular markers and extravasate into the tumour microenvironment through the enhanced pore and retention effect (EPR) [3].

droplets [10]. This literature review however will specifically focus on the nanodroplets which are capable of phase-change from a liquid to gaseous state. Readers interested in the other named nanoparticle agents can be referred to other reviews by Sirsi and Borden [14] as well as Cavalli, Bisazza & Lembo [15].

2. Physics of ultrasound waves

The physical phenomenon of wave generation begins with the inverse Piezoelectric Effect where an electric pulse is applied to energise the piezoelectric crystals in the ultrasound transducer [16]. This changes the alignment of dipole moments inside, resulting in mechanical pressure waves travelling out of the probe and into the patient's body [17]. The Piezoelectric Effect then ensues some time later where echoes are returned to the transducer to be converted back into electrical pulses, translated into pixels and essentially create images on the monitor [1]. Such a procedure is repeated thousands of times per second to create two dimensional 'real-time' images [17].

Whilst the energy is propagating through the medium, it generates alternating areas of compression (area of high density) and rarefaction (area of low density) whereby the magnitude of these depends on the amount of applied acoustic energy [18]. These two processes are demonstrated in Fig. 2 where compression is represented by the darker shade, and rarefaction is indicated by the white area in response to the time-varying acoustic pressures of the ultrasound field [19].

2.1. Acoustic cavitation and sonoporation

The fluctuating compression and rarefaction processes of the sound wave exhibited by its growth and collapse of gas pockets is termed acoustic cavitation (AC) [21]. This fact is used to the advantage of UCA to promote more efficient and targeted drug delivery by increasing sonoporation for drug uptake [14]. Sonoporation is the modification of vascular and cellular membrane permeability through the formation of temporary pores [20]. The pores act as a filter in which the size of the opening determines the extravasation or internalization of UCAs [3]. AC also produces strong backscatter echoes of the ultrasound waves and

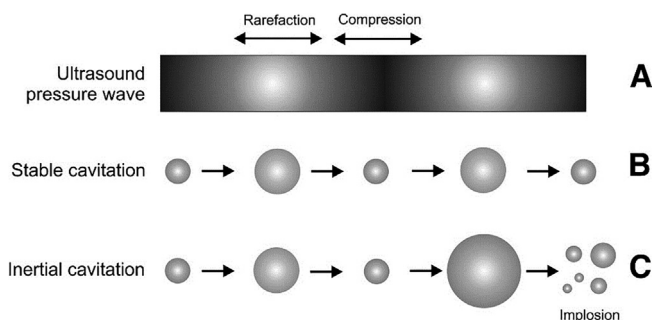


Fig. 2. (a) The changes in an ultrasound wave as it undergoes compression and rarefaction. (b) and (c) are schematic representations of an acoustic pressure wave differentiating stable and inertial cavitation of microbubbles respectively. Of note is the implosion of the microbubble into many smaller fragments in (c) because it results in much stronger biophysical effects [20].

hence intensifies the reflected signal [22].

AC is categorised into two forms- stable or inertial cavitation depending on the intensity, amplitude and frequency of the ultrasound wave used, as well as the size and composition of the contrast agent [21]. The former transpires at low amplitudes as sustained, linear oscillations generating micro streaming in the surrounding fluid [23]. The latter occurs as larger oscillations at higher ultrasound intensities causing the microbubble to grow non-linearly until it reaches its maximum size and eventually collapses into many smaller microbubbles [24]. This fashions several violent phenomena, such as microjets and shock waves [22]. If this collapse of the microbubble occurs close to a cell membrane, they are able to puncture the surface and permeabilise blood vessels due to the strength of the microjets and shockwaves [25]. Cell membrane perforation and consequent creation of pores can benefit drug delivery to cells [26]. Despite inertial cavitation promoting sonoporation for increased cell membrane porosity to drugs, genes and their carriers, it can cause some of the mechanical and biological effects of ultrasound [7]. Fig. 3 demonstrates four different mechanisms of sonoporation.

However, once the ultrasound signal has been removed, cell membrane permeabilization swiftly decays and the pores re-seal as rapidly as milliseconds to seconds [27]. This indicates that the pores only exist as long as the oscillating microbubbles and ultrasound field are present and thereafter, cell membrane potential is restored [28].

2.2. The mechanical index

As described above, the force of the ultrasound pressure wave from the cavitation processes can be so powerful that the microbubbles grow so quickly that they break into fragments [31]. To account for the possible bioeffects of this implosion and disruption to molecules, the mechanical index (MI) was established. The MI is displayed on ultrasound scanners to provide a measure about the magnitude of energy of an ultrasound beam and acts as an indicator of potential cavitation [32]. It is given by Eq. (1) where P_{AC} is the peak rarefactional acoustic pressure in mega Pascals, MPa, and f is the centre frequency in mega Hertz, MHz of the ultrasound beam [25].

$$MI = \frac{P_{AC}}{\sqrt{f}} \quad (1)$$

It is imperative to note that the MI does not perfectly predict biological consequences as it does not take into account the existence of ultrasound-sensitive materials in the body [14]. As specified by the Food and Drug Administration, the maximum MI for diagnostic imaging to avoid bioeffects is 1.9 [33]. This threshold value however only applies when imaging with micelles or liposomes which are incompressible drug carriers. Since microbubbles interact strongly with ultrasound, the upper MI limit is lowered to 0.8 [14]. The reason for

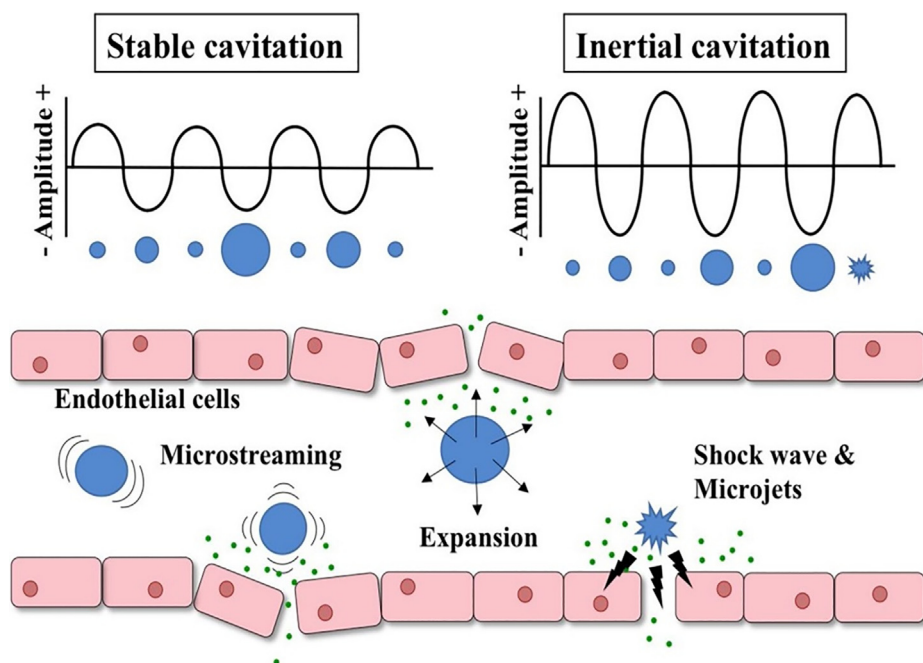


Fig. 3. Biophysical effects of AC on endothelial cells. Stable cavitation occurs at low amplitudes where the microbubbles (MBs) oscillate linearly and create acoustic micro-streaming. Increased MB size can cause invagination and membrane opening. Inertial cavitation occurs at higher amplitudes where the MBs oscillate non-linearly. This results in expansion of the MB, pushing the membrane apart, ultimately disturbing membrane integrity and prompting MB collapse releasing shockwaves and micro jets creating openings in the endothelial wall [29].

this reduction is that at lower MIs, specifically below 0.5, microbubbles undergo stable cavitation where no growth occurs, however the vibration is enough to produce the ultrasound signal [34]. However, in order for droplets to transition to bubbles, higher intensities are required. The higher MIs increase contrast to noise ratio between blood and tissue from the bursting of the microbubbles because the theoretical risk of cavitation increases [35]. This is due to non-linear oscillations eventually instigating the microbubble rupture and this is also responsible for the non-thermal bioeffects related to acoustic cavitation [32].

Despite MI being reliant on acoustic intensity, there is no direct association between the MI or microbubble destruction as it is unable to forecast the scope of such a process. This means the MI should not be used solely to set the exposure conditions for the demolition of microbubbles [36]. In saying this, there is still a demand for an official recommended maximum MI when imaging with phase-change agents [14].

2.3. Nanodroplets

The majority of reported and tested contrast agents are microbubbles due to their many advantages. These include a low acoustic impedance (~ 0.3 MRayl), therefore higher echogenicity, which consequently provides better contrast [10]. In addition, microbubbles dramatically increase intracellular uptake of drugs or genes as they undergo inertial cavitation [24]. Drug delivery is thus enhanced through this growth and collapse that concentrates ultrasound energy [37]. Unfortunately, microbubbles cannot be selectively placed in tumour tissue to perform targeted diagnosis and therapy as they are too large of a microscopic dimension to pass through vascular endothelium and into extravascular space [38]. Their clinical applications are accordingly restricted to intravascular areas and are predominantly used to detect blood perfusion and velocity [5].

A resolution to this was the inception of phase-change nanodroplets which are of the submicron size at 400–800 nm. This small size enables effective extravasation beyond the vasculature to directly target disease bio-markers in tumour tissue [3]. They are PFC based encapsulated in a phospholipid, protein, surfactant or polymer-based shell [12,15]. These contrast agents are stable, capable of remaining in the nanoscale and drug loaded to be used in targeted molecular imaging [10].

2.4. Acoustic droplet vaporisation (ADV)

These specially designed PFC droplets have a significant feature in that they are able to maintain their shape and original diameter upon initial injection [3]. This allows nanodroplets to extravasate and accumulate at the site of interest before hypobaric activation causes the liquid droplets to undergo a phase conversion into microbubbles [4]. The phase transition occurs through a physical process known as acoustic droplet vaporisation (ADV) in which heating provoked by ultrasound and ultrasound pressure waves induce the conversion from liquid nanodroplets to gaseous microbubbles [39] (Fig. 4). This phase shift is necessary because droplets have a similar acoustic impedance to surrounding tissue and therefore have an inherently low contrast. However, once these droplets are converted to gas bubbles, their echogenicity increases, and they can effectively be used as contrast enhancers [40].

Rapoport and co-workers were one of the first to produce this alternative class of ultrasound contrast agents and this paper explains the generation of such PFC nanoparticle drug carrier [41]. The first generation developed were made using amphiphilic doxorubicin-containing block copolymer micelles that were stabilised by an echogenic PFC, namely perfluoropentane (PFP). Such a formulation was coated with polylactic acid- polyethylene glycol (PLA-PEG) [39]. A mixture of these nanoemulsions were created, ranging from 200 to 750 nm, depending on the kind and ratio of copolymer to PFP. Their variations were able to be drug-loaded and were made of poly (ethylene oxide) – co-poly (L-lactide) (PEG-PLLA), poly (ethylene oxide) – co-poly (D, L-lactide) (PEG-PDLA), or poly (ethylene oxide-co-polycaprolactone) (PEG-PCL) micelles. The functions that these contrast agents could perform were drug carrying for delivery to the intracellular layer, tumour targeting and enhancing the ultrasound contrast of tumours [10].

2.5. Laplace pressure

While reducing contrast agents from the micro to nanometre size increases their capacity to accumulate in tumour tissue and lengthens their circulation time in the body, pitfalls have nevertheless been identified in that there is diminished echogenicity and stability of the bubbles in solution [42]. Bubble stability is contingent on the opposing forces of the partial pressures of dissolved gasses in the surrounding

(b) Phase-change Droplets (PCDs)

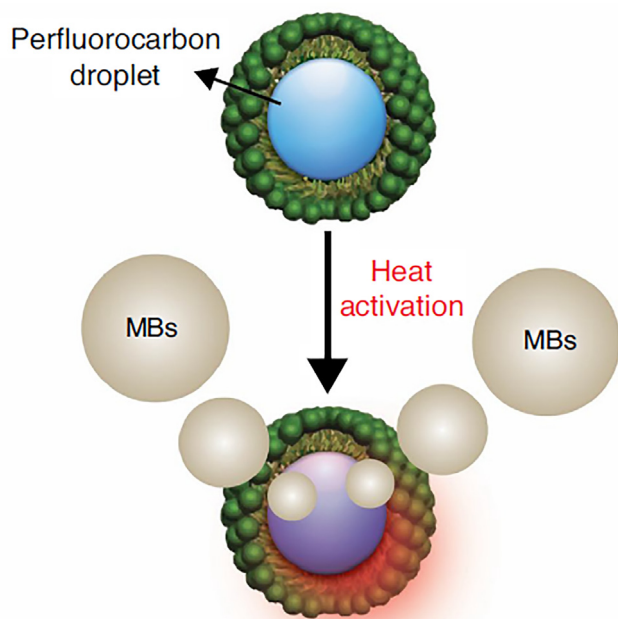


Fig. 4. Schematic figure of PFC droplets encapsulated in phospholipids or polymers phase changing into MBs upon heat activation [3].

fluid and contributions from both the Laplace and ambient pressures [43]. The Laplace Pressure is given by Eq. (2) where P_{inside} is the pressure inside a droplet, P_{outside} is the pressure outside a droplet, σ is the surface tension at gas–bubble interface, and r is the droplet radius [44].

$$\Delta P_{\text{Laplace}} = P_{\text{inside}} - P_{\text{outside}} = \frac{2\sigma}{r} \quad (2)$$

A qualitative description of the Laplace phenomenon is that it contributes additional pressure to the interior fluid of a droplet. This is generated by the surface tension or interfacial energy at the interface of two dissimilar phases. Because the molecules of each of these immiscible phases are not the same, a force is created and acts as a membrane between the two—the liquid and gas [39]. This has important consequences for drug delivery as it influences the droplet's threshold of vaporisation and succeeding volumetric expansion [10].

The surface tension between liquid PFCs and air is low, however due to the hydrophobic nature of the PFCs, when dispersed in water, this elicits high interfacial energy [45]. As demonstrated in the second graph of Fig. 5, the rarefactional acoustic pressure phase is called sub-pressurisation. During this rarefactional stage of the ultrasound cycle, the sum of the internal pressure of the PFC droplets and the Laplace pressure is less than the vapour pressure of the PFC liquid [46]. This difference permits the liquid to convert into a gas however at larger acoustic pressures, it condenses back into a liquid [47]. As indicated by the equation, Laplace Pressure is inversely proportional to droplet radius, and thus smaller droplets have higher boiling temperatures, greater Laplace Pressure and are less easily vaporised into bubbles at lower pressure than larger droplets [10]. During the growth and collapse of bubbles under the influence of an acoustic field, the bubbles may also undergo fragmentation and coalescence. When bubbles start to coalesce, the probability of bubble survival is significantly higher for coalesced bubbles than non-coalesced bubbles due to the increase in total volume and size for the bubbles as well as the reduced Laplace pressure [47]. Apart from bubble coalescence, influx of dissolved gases from surrounding medium and diffusion of PFC from small bubbles to

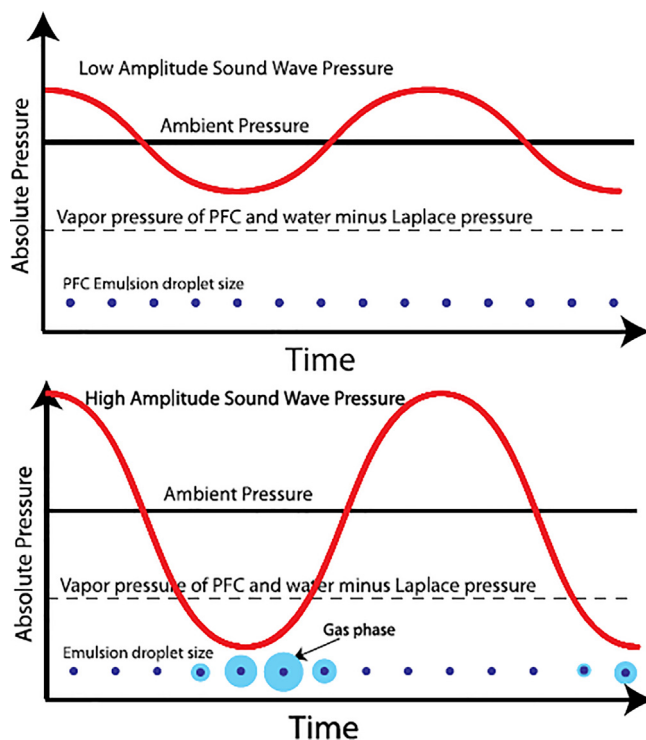


Fig. 5. Schematic plot of an ultrasonic acoustic wave in a PFC. The top graph shows an acoustic wave with low amplitude pressure above the difference between the vapour pressure of the PFC and the Laplace Pressure. The bottom graph shows the vaporisation of the liquid droplet into a gas bubble as soon as the local acoustic pressure of the wave drops below the pressure difference between the vapour pressure of the PFC and the Laplace pressure (not to scale) [46].

larger bubbles can occur and result in bubble growth and impede reversible vaporisation. Thus bubble expansion will not follow the ideal gas law [48].

According to Henry's law, the solubility of gases increases with pressure. Therefore, during the rarefaction stage of the ultrasound pressure wave, the droplet-to-bubble transition into nanodroplets is aided by the high solubility of gases. Oxygen for example, in the gas phase is highly dissolvable, which has allowed the use of perfluorocarbon emulsions as blood substitutes. On the other hand, in the compression stage, some gases may be released. However, the degree of release is less than the uptake due to high solubility under pressure and smaller surface area of compressed bubbles making the effects negligible. Upon removing the ultrasound signal, equilibrium is restored whereby gases with super-equilibrium concentrations diffuse out and thus restore nanodroplets. Since non-condensable gas may not completely dissolve into surrounding liquid, a small bubble composed of this gas will be left to nucleate in the next cycle of PFC boiling. This produces an even larger bubble which correspondingly results in more diffusion of non-condensable gas into the bubble [41].

The vapor pressure of water is 3.17 kPa at 25 °C, an order lower than that of a small PFC (i.e., 29.1 kPa for PFC6). Thus, PFC bubbles will form before homogeneous nucleation of water vapor bubbles. However, in practice, a small portion of water evaporates into the first bubbles to form. The first perfluoropentane bubble to form contains 3.5 and 4.4% water vapor in equilibrium at 25 and 37 °C, respectively. Perfluorohexane has a lower vapor pressure, so the corresponding values are slightly higher, 9.8 and 11.6% at 25 and 37 °C, respectively [33]. The complexity in the vapor components would lead to complicated bubble dynamics after vaporization [47].

Another determining factor in the transition from a liquid to gas is time, which is proportional to the probability of a nucleation event and

exponentially related to the magnitude of sub- pressurisation [49]. Thus, even if the vapour pressure is at a temperature above the boiling point and there is adequate Laplace pressure, if it is not greater than the pressure inside the droplet, it will prevent gas formation. This is called apparent superheating [50].

As such, the Laplace pressure is only experienced by the PFC droplets beyond ambient pressure when surface tension effects outweigh the radius and the boiling temperature has been reached [51]. This is evidenced by PFC nanodroplets experiencing Laplace pressures several atmospheres above ambient pressure and still not vaporising because the boiling point has not been met [51]. Nonetheless, a PFC with a low boiling point of 29 °C is PFP and it is still able to form into echogenic bubbles of 1–2 µm using lower acoustic amplitude [2].

Another issue to consider is how big the PFC droplet will be once it is vaporised. Features that determine the extent of the expansion are initial diameter of the droplet, ambient pressure and density. Relating back to the Laplace equation, since the Laplace pressure is inversely related to radius, the effects are more noticeable for droplets in the nanometre range [51]. This means that such nanoscale bubbles will experience greater Laplace pressure than droplets at the microscale and the gas core will be more compressed [52]. It can thus be deduced that for smaller droplet sizes, their expansion occurs less readily than droplets in the micrometre range attributable to the effect of increased Laplace pressure [51].

Regardless of the amount of expansion, the newly formed bubbles stably oscillate for a number of cycles before shrinking or collapsing [47]. The fate of vaporised bubbles is not well known however Reznik *et al.* have proposed that a 500 nm gas bubble has a Laplace Pressure of 300 kPa. Also accounting for the hydrostatic pressure of 100 kPa, the total pressure inside the bubble is thus 400 kPa. This pressure is enough to return the bubble back into the liquid droplet form [52]. Since the Laplace Pressure increases with decreasing radius this reduces the stability of the bubble, hence research has been performed to stabilise the PFCs. One such method is coating the droplets with surfactant molecules at the interface between the gas bubble and surrounding liquid environment. When the droplets are not coated, the surface tension of a dodecafluoropentane droplet is found to be 56 ± 1 mN/m. However, when the droplets are coated with poly (ethylene oxide) (PEG) shell, this reduces the effective interfacial surface tension of the bubble to 30 and 50 mN/m as displayed in Fig. 6 [10].

This encapsulation of droplets with a lipid or polymer shell consequently reduces the Laplace Pressure and internal bubble pressure so that when the liquid droplet is vaporised by ultrasonic radiation, it will be a large and stable bubble [24]. This increased stability will reduce the contribution of the Laplace Pressure, effectively preventing

recondensation back to the liquid state due to the static and dynamic molecular properties now present across the interface [53]. The relationship between concentration of surfactant coating and likelihood of condensation can be described by the vaporised droplets with a high concentration of surfactant shell material being less likely to condense back into a liquid than those with less surfactant [52]. The increased bubble survival is attributed to the increased total volume and size of the bubble. As such, even if bubbles are of similar size, if there is variation in the shell material on each bubble, such as longer PEG chains or denser coating, the transition back into a liquid will be different because stability will differ [51].

2.6. Nanodroplet formulations

As discussed earlier, the effects of surface tension on nanodroplets can be minimised with the addition of stabilising surfactants or lipids. However, the selection of the core is also crucial to its durability so that circulatory retention time can be long enough to meet the needs of clinical imaging [54]. For liquid droplets to be able to circulate in the body before being evaporated into a gas, the liquid needs to be non-toxic and immiscible with water [39]. Thus, unlike microbubbles which are normally formed using air, nitrogen or sulphur hexafluoride as the core, nanodroplets use PFC gas as the core to meet these criteria [55].

PFCs are high molecular weight gases that are chemically and biologically inert in that they perfuse less easily and are less diffusible than air. Their low solubility in blood also leads to decreased bubble dissolution time [4]. As such, the PFC most commonly used in these ADV creations is PFP due to its high vapour pressure, price, accessibility and relatively low toxicity [39]. Other traits of PFP droplets is that its boiling point of 29 °C means that droplets of 1–5 µm in size remain in a superheated state whilst they circulate within the body. It is only when a minimum threshold pressure has been achieved that they are vaporised to bubbles by ultrasound irradiation [55].

Depending on the application that the phase-change nanodroplets will be used for- such as drug carriers for tumour targeting and detecting cancer, they must have appropriate characteristics [56]. These include stability in circulation for adequate circulation time, drug retention until activated, size that facilitates extravasation through defective tumour vasculature and ultrasound responsiveness in order to undergo ADV [10]. Factors that affect the circulation time, biological compatibility and stability will be further explored.

2.7. Circulation time

Unlike microbubbles which have a circulation time of minutes, nanodroplets have a half- life in the order of 2–6 h for imaging applications. For therapeutic use, the half- life extends to several days, which allows the particles to accumulate at the intended targeted [57]. Methods to lengthen circulation period involves ensuring stability to avoid reticuloendothelial system (RES) uptake. If not protected, the body's RES will clear the foreign nanodroplets from the circulation before reaching the target site [15].

Another tool to elongate circulation time of nanodroplets is by modifying its surface to alter environmental exchanges. This is done by incorporating polyethylene glycol (PEG) to the surface of nanodroplets to act as a steric hinderer and pressure resistor [43]. Moreover, the PEG coating behaves as a hydrophilic protector by blocking and delaying the interaction between blood or protein [54]. Diaz-Lopez *et al.* performed a study to trial this and found that the pegylated nanodroplets can effectively extravasate in tumours due to the extended time in the target area as it reduces the possibility of recognition by phagocytic cells to engulf them [58]. This prolonged circulation time as a liquid before being triggered by ultrasound, caused by shielding from opsonin absorption and discharge by the RES is beneficial to the passive targeting of tumour tissue by the EPR [59].

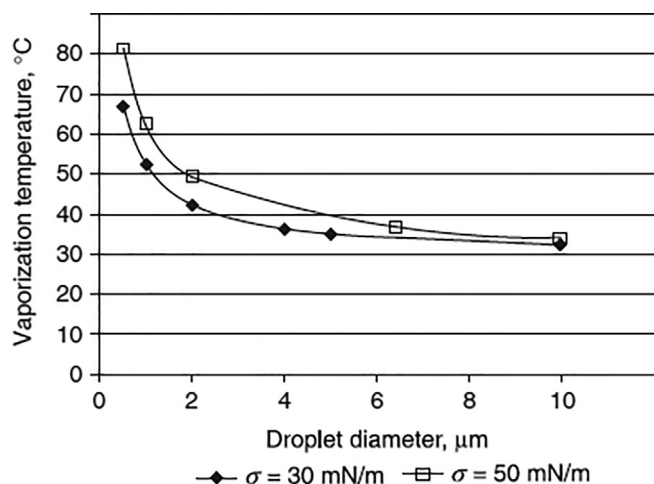


Fig. 6. Droplet diameter versus vaporisation temperature for two dodecafluoropentane droplets of differing surface tension, 30 and 50 mN/m [10].

2.8. Biological compatibility

As already stated, one of the principle uses of contrast agents is drug delivery. Ultrasound mediated drug release is prompted by the physical effects of cavitation resulting from the ultrasound wave propagating through tissue and experiencing compression and rarefaction, as mentioned previously [18]. Thus, nanodroplet drug carriers are designed specifically to respond to this ultrasound trigger. The performance of the PFC in drug delivery is measured by the therapeutic index (TI) which is defined in Eq. (3).

$$TI = \frac{TD_{50}}{ED_{50}} \quad (3)$$

TD_{50} is the drug dose that produces a toxicity in 50% of the population and ED_{50} is the minimum effective dose for 50% of the population [14].

A high therapeutic index is optimum to increase the targeting potential whilst simultaneously giving appropriate dose required for toxicity and decreasing that for effective therapy. By doing this, the majority of drug is delivered to target tissue and healthy tissue is avoided [14].

2.9. Combining PFCs

Alternatively, combining PFCs can also increase stability as demonstrated by a study performed by Phillips et al. They created a novel phase-shift nano agent by combining highly volatile decafluorobutane (DFB) with less volatile dodecafluoropentane (DDFP). By using both in a 1:1 ratio, the goal was to balance stability with acoustic sensitivity. Indeed, their study found that vaporisation and ablation were achieved at relatively low high intensity focused ultrasound (HIFU) intensities and droplets were still stable enough to remain in the liquid phase for up to 48 h [60]. Moreover, the droplet sizes were as small as 100–300 nm which enables extravasation in tumour tissue [14].

Other researchers have also performed studies to improve the efficiency of droplet to bubble transition by using different mixtures. For example, a study by Kawabata et al. aimed to see if controllable phase-shift ultrasound contrast agents could be developed by using emulsions containing multiple PFCs [61]. They combined decafluoropentane (DFP) and PFP and found that the two required different ultrasound intensities to be vaporised. This is due to the two liquids having different boiling points, DFP = 55 °C and PFP = 29 °C [44]. As supported by Rapoport et al., ADV is considerably catalysed by large pre-existing bubbles irradiated by low-frequency ultrasound because they do not need to be as sensitive to ultrasound, pressure or temperature as agents used in conventional formulations [62].

3. Applications

Phase-change PFC nanodroplets have great promises in a wide range of clinical and therapeutic applications due to their desirable physical properties [40]. Such applications include targeted drug carriers [24], embolotherapy [63], gene delivery [64], enhanced thermal ablation [51], phase aberration correction [65] and many more. For the purposes of this review only drug delivery and embolotherapy will be broadly discussed in terms of cancer therapy.

Despite microbubbles showing certain characteristics in medical use, phase-change nanodroplets have more favourable traits such as better controlled release of drugs and lower toxic effects [44]. They are introduced in the liquid state on the nanoscale and are designed to exploit the enhanced permeability of tumours by extravasating and accumulating into tissue through leaky vasculature [66]. They can circulate stably for hours however once exposed to ultrasound, they become acoustically activated, expand and form vaporised gaseous microbubbles. The increase in volume means the newly formed bubbles are now able to target tumours to deliver drugs or occlude vessels for

cancer treatment [17,67]. The droplet vaporisation is dependent on local rarefactional pressure and thus microbubbles will only be generated within the narrow focal region where acoustic properties are the strongest, guaranteeing that the therapeutic effect is limited to the desired treatment areas [68].

3.1. Drug delivery

Current traditional chemotherapeutic agents such as doxorubicin and cisplatin are not favourable due to their low specificity in being able to distinguish healthy and tumour tissue and thus, they have a narrow window of therapeutic efficacy. The drugs attack all cells without discrimination, being especially harmful to any rapidly multiplying cells such as hair, intestinal epithelial cells, and bone marrow [14].

In order to reduce the severe side effects chemotherapy can inflict, drug-loaded ultrasound contrast agent systems, also termed ultrasound-responsive drug delivery systems (URDDS) have been designed. These nanodroplets contain drugs in their cores such as chemotherapeutic medications or hydrophilic compounds [9]. Through acoustic cavitation, high ultrasonic pressure induces violent oscillation of the ultrasound contrast agent, leading eventually to their obliteration. This ultrasound stimulus triggers the bubbles to be active for enhanced drug delivery at the desired sites [54].

The advantage of using nanoscale carrier systems is their specificity in targeting diseases. The drug deposition is performed with high spatial precision because the mechanisms of interaction between drug carriers and ultrasound waves promotes drug release and tissue uptake [11]. Thus, the therapeutic agents are administered selectively to targeted disease sites without adverse effects on healthy tissue [59]. As such, nanoparticles are potent drug carriers for chemotherapeutic agents, have the aptitude to drastically improve the way cancer is treated and can offer better diagnostic methods by alleviating some of the cytotoxic effects to healthy cells caused by conventional chemotherapeutic agents [69].

In a study by Lee et al., chitosan-deoxycholic acid coated nanodroplets were made containing PFP as the liquid core (Fig. 7a). Small interfering RNA (siRNA) were electrostatically bound to the surface to test its delivery to lung and breast cancer cells. They demonstrated that their droplets could undergo ADV with using the parameters 1.8 MHz and 335 kPa peak negative pressure (Fig. 7b) and the passive acoustic mapping (PAM) results showed that there is an enhancement in the acoustic signal when chitosan nanodroplets were used compared to the control samples (cells only and solid chitosan particles) (Fig. 7c-f). Overall, their study found that 72 h after ultrasound delivery of siRNA, there was significant cell death (52.4%) without impairment of functionality. As shown in Fig. 7g the combination of the nanodroplets, siRNA and ultrasound were the most effecting in the number of cells killed. Thus, siRNA has substantial therapeutic potential in targeted drug delivery for the treatment of cancer [70].

3.2. Embolotherapy

Phase-change droplets also find application in embolotherapy. Gas embolotherapy is a novel technique for cancer treatment in which targeted vessels are occluded deliberately to reduce or stop regional tumour blood flow [71]. The PFC droplets flow through the vasculature feeding a tumour until gas microbubbles are selectively formed by ADV upstream of intended occlusion sites [63]. The microbubbles then flow downstream and occlude vessels which have diameters sufficiently smaller than the microbubble diameter [71]. The resulting bubbles can persist for hours to days, appropriately occluding the tumour tissue from blood flow. This induces cell death, producing ischaemic damage and effectively shrinking the tumour. This is displayed in Fig. 8 [72].

Compared to conventional embolization techniques, gas embolotherapy offers some advantages. These include being less invasive,

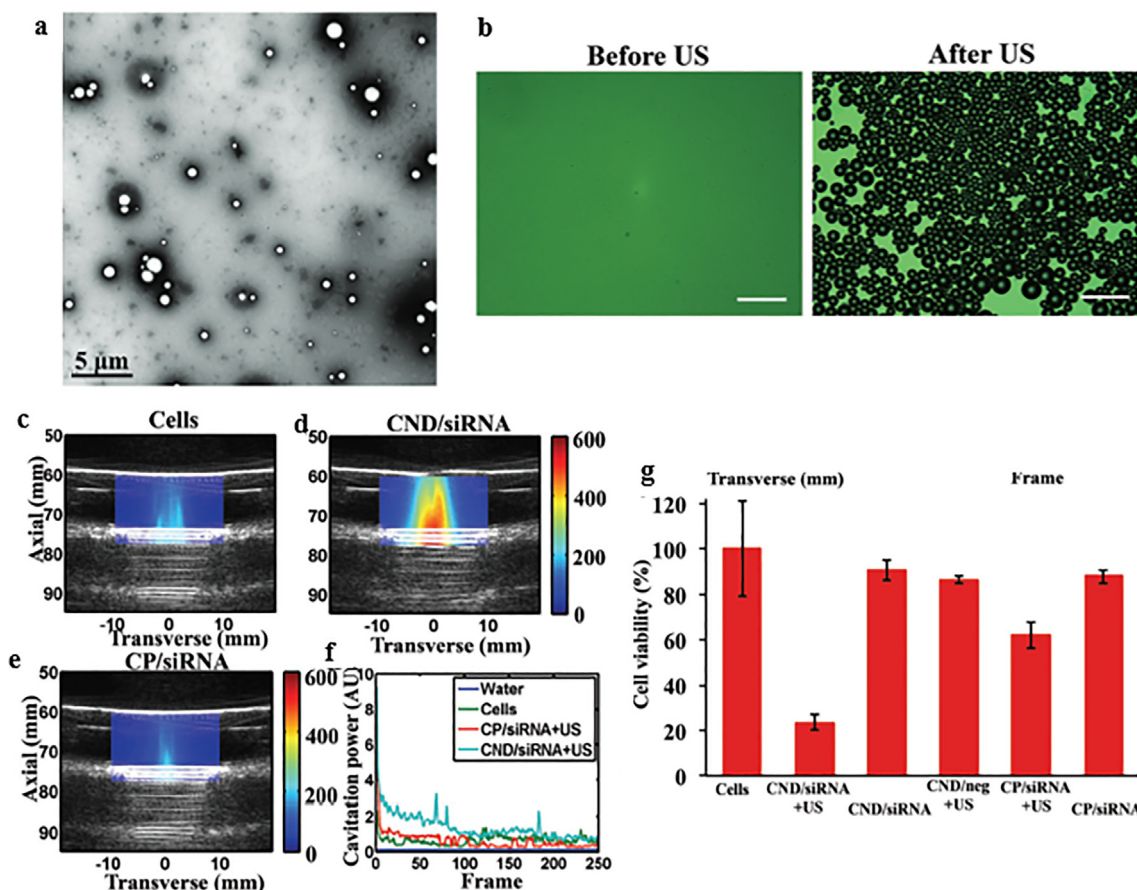


Fig. 7. (a) Transmission electron microscopy (TEM) image of the nanodroplets before ultrasound exposure. The white spots are the PFP core and the darker rings are the chitosan-deoxycholic acid coatings. (b) Optical microscope images of the nanodroplets before and after ultrasound exposure for 45 s. The nanodroplets were easily vaporised into bubbles following ultrasound exposure. (c–f) Passive acoustic maps of summed cavitation energy overlaid on B- mode images showing the evolution of microbubbles during ultrasound exposure. (c) Cells only and no enhancement in signal, (d) CND/siRNA showing a slight increase in signal enhancement, (e) CP/siRNA. Colour bars represent cavitation power (AU). (f) Maximum cavitation energy (AU) from each frame of acoustic mapping data. (g) Percentage of cell viability exposed to ultrasound and incubated for 72 h with various mediums whereby the combination of the nanodroplets, siRNA and ultrasound were the most effecting as it killed the most cells [70].

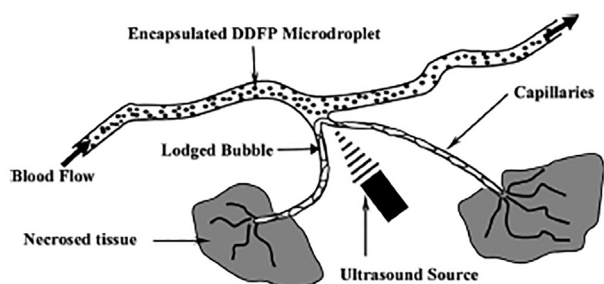


Fig. 1 Schematics of the gas embolotherapy procedure

Fig. 8. Schematic of the gas embolotherapy procedure showing microdroplets travelling in blood. Upon high- intensity ultrasound, ADV is initiated forming microbubbles near the desired occlusion site. These block blood flow and the tumour tissue becomes necrotic [72].

having more accuracy, presenting lower risk of infarction of healthy tissues, and facilitating drug delivery to targeted sites [63]. This is because vaporisation can be induced intentionally and accurately in the feeder arteries of tumours, resulting in occlusion with a high degree of spatial specificity [39].

Currently, gas embolotherapy has been demonstrated experimentally for staging and prognosis of hepatocellular and renal carcinoma

however ADV induced embolotherapy also has great potential in crucial organs such as the brain, pancreas and prostate [63]. All studies for vessel occlusion have been performed in rodents or dogs thus are yet to be translated to clinical use [47]. One of the reasons for this is caution being required in carrying out embolotherapy because many arterial emboli could create infarcts in the heart or brain, or travel to distant vascular beds where they could cause unwanted arterial occlusion, leading to obstruction and ultimately ischemia [73]. Moreover, factors that determine the performance of ADV-based occlusion includes interaction with vessel bifurcations and how easily the bubbles will coalesce into sufficiently large ones to occlude arterioles and capillaries, are still under investigation [47].

3.3. Unsolved problems and future perspectives

As presented in this literature review, there have indeed been a lot of studies on phase-change nanodroplets that have shown good prospects in a range of clinical applications. However, there now needs to be a focus on how to best formulate such agents to optimise their performance in clinical applications, as well as to have a better understanding of ADV and the parameters that affect it [47].

One of the difficulties in designing phase shift nanodroplets is ensuring that the droplets change phase under ultrasound exposure but remain insensitive to temperature or pressure changes occurring inside the human body [44]. The issue is that phase change depends on these

two factors. Another critical problem relating to ADV is the Laplace pressure. As denoted by Eq. (2), the smaller the droplet, the more acoustic pressure and high temperatures are required to stimulate vaporisation which can harm surrounding tissues [10]. A solution to this is mixing PFCs to better control the vaporisation threshold however further investigations are needed to determine the best combination of PFCs as well as to determine the exact threshold that inspires ADV [61].

Moreover, the ADV mechanism and parameters that influence the dynamics of bubble evolution are not yet fully understood [74]. Parameters that need to be investigated are the determination of optimal initial droplet size, the correct acoustic factors such as frequency, peak pressure and duty cycle that will achieve the desired bubble size, how to ensure that all droplets vaporise into gas, and that ADV occurs at the required location [63]. A particular issue is that upon vaporisation, the expansion is hard to control thus some bubbles may become too large due to the influx of dissolved gases present in surrounding media [51].

As such, the existing challenges in controlling the onset and rate of vaporization, microbubble size and subsequent coalescence have complicated clinical translation. Therefore, studies to assess the aforementioned clinical uses of phase-change droplets have been done in vitro however they are yet to be trialled on humans [14].

4. Conclusion

This review has outlined the historical development of phase-change nanodroplets, their physical processes such as acoustic cavitation, Laplace Pressure and acoustic droplet vaporisation and how their various formulations and unique properties can be used in a spectrum of therapeutic applications. They can be engineered for specific uses, are able to access vasculature and most organs of the body, and limited side effects and long-term impairments have been revealed in the literature. Nonetheless, a need for further comprehension in the ADV phenomena and its parameters have been identified. This is in regards to the inability to control the rate and magnitude of vaporisation and the resultant bioeffects. Owing to the promise of safe and cost-effective ultrasound imaging that might be possible with phase-change nanodroplets, more clinical trials need to be pursued to ascertain whether they can indeed be a valuable tool for diagnosis and treatment.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ultrasonch.2019.03.024>.

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